Extracting Predictability: A Time-Varying Factor-Augmented Approach under Mixed Persistence

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Abstract: This paper develops a time-varying factor-augmented forecasting model designed to extract predictable variation in the target variable when predictors exhibit mixed persistence. The framework accommodates stationary, nonstationary, and a mixture of observable and latent predictors, and allows latent predictors to follow either static or dynamic factor structures, thereby nesting many classical and new models as special cases. We develop an easy-to-implement sieve estimation procedure for the time-varying coefficients, establish its large-sample properties, and construct prediction intervals for the forecasts. Simulation studies show that the proposed methods perform well in finite samples. In an empirical application to U.S. inflation forecasting, the proposed model performs favorably relative to several competitors. Further analyses indicate that latent factors remain informative even when inflation exhibits autoregressive dynamics, since they absorb predictable variation that would otherwise be left in the autoregressive model errors.

Keywords: Factor-augmented regressions, Time-varying coefficients, Sieve method, Nonstationary factors, Dynamic factor models

JEL classification: C32; C38; C53

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1 Introduction

Forecasting economic variables, i.e., modeling their conditional expectations given an information set, is essential for policy formulation, financial decision-making, and macroeconomic risk assessment. As Jurado et al. (2015) argue, forecasting models must be sufficiently rich to capture all predictable components of future outcomes; otherwise, the model residuals will contain predictive information, leading to biased forecasts. Factor models play a crucial role in addressing this challenge. By extracting latent factors from high-dimensional data, factor-augmented regression (FAR) provides a powerful forecasting framework in the presence of many potential predictors (see, e.g., Stock and Watson 2002*a*; Bai and Ng 2006; Cheng and Hansen 2015, and the references therein). Its practical merits have been demonstrated in a wide range of applications, including measuring economic uncertainty (Jurado et al. 2015), forecasting macroeconomic variables (McCracken and Ng 2016), and predicting financial returns (Bauer and Hamilton 2018; Dong et al. 2022). Yet, several empirically relevant features may prevent conventional models from exhausting this predictability, leaving predictable information in the residuals.

One important source of such residual predictability is structural instability in forecasting relationships (e.g., Rossi 2021). Driven by policy shifts, technological advances and global shocks, the unconditional means and variances of economic variables can exhibit substantial instability over economic cycles (e.g., Stock and Watson 2002*b*, 2005), so that fixed-parameter models such as traditional autoregressive (AR) models tend to produce larger forecast errors, leaving predictable variation in the residuals. Another source arises from the mixed persistence of economic predictors. For example, inflation forecasts typically combine persistent cost-side indicators such as commodity prices, energy prices, and unit labor costs with stationary measures of cyclical slack and financial conditions, such as the output gap, unemployment gap, and interest-rate spreads (e.g., Faust and Wright 2013; Medeiros et al. 2021), while equity-return forecasts mix persistent valuation ratios (e.g., the log price-dividend ratio) with stationary risk-premium proxies such as term and credit spreads (Goyal et al. 2024). Existing forecasting regressions, however, typically address such persistence heterogeneity by differencing the nonstationary variables before estimation, which can discard low-frequency predictive information and weaken forecasting performance (e.g., Choi and Jeong 2020).

So far, these two issues have been studied extensively but mostly in isolation: structural instability through time-varying models (e.g., Cai 2007; Gao et al. 2024*a*,*b*), and mixed persistence

through cointegration regressions (e.g., Park and Phillips 2001; Phillips and Wang 2023; Gao et al. 2025). Nevertheless, whether parameter instability and nonstationarity are studied separately or jointly, the existing literature is largely restricted to models with observed covariates. A flexible forecasting framework that can simultaneously handle such features while accommodating both observable and latent predictors has yet to be developed, despite its clear importance and the urgent need for it in economic forecasting.

To tackle the above issues and meet the practical needs of economic forecasting, we propose a general regression model that accommodates time-varying coefficients, a nonstationary factor structure, and autoregressive dynamics. The framework is flexible in that, through a time-varying coefficient specification, it allows for observable predictors, latent factors, or both, with components that may be stationary, nonstationary, or a mixture of both. The flexibility also extends to such latent factors that may follow either static or dynamic structures. As a result, the proposed framework nests a wide range of extant and new models as special cases, including time-varying coefficient models (Cai 2007), time-varying factor-augmented forecasting models (Wei and Zhang 2020), time-varying cointegrating regressions (Phillips et al. 2017), and factor-augmented regressions with mixed stationary/nonstationary factors (Hannadige et al. 2024).

In this paper, we first propose an easy-to-implement estimation procedure for the model, in which the time-varying coefficient functions are estimated by a sieve method, and the latent factors, when present, are recovered by principal component analysis. We then establish the asymptotic distribution of the proposed sieve estimators, with convergence rates that depend on the persistence of the regressors. Assuming normally distributed errors in the forecasting model, we further show that the forecast error is asymptotically normal, which allows us to construct valid prediction intervals for the target variable under consideration. Monte Carlo evidence reported in the online supplementary material demonstrate desirable finite-sample performance across various data-generating processes.

In an empirical application, we apply the proposed model to forecast U.S. inflation. The results show that, under the most general specification, the proposed model compares favorably with its natural competitors, owing to its ability to capture structural instability and to exploit information from both persistent and transitory factors. We further show that autoregressive persistence does not exhaust the predictable component of inflation: latent factors remain informative because they absorb low-frequency and cyclical information that would otherwise be left in the forecast

error. The economic content of these factors is transparent: the extracted nonstationary factor captures permanent shifts in the nominal anchor and serves as a latent proxy for long-run inflation expectations, while the stationary factor isolates mean-reverting cyclical pressures and serves as an empirical proxy for the output gap.

Our paper contributes to two related strands of literature while addressing important gaps in both. First, it is related to a diverse body of research that addresses structural instability in factor-augmented frameworks, where parameter change is captured through various modeling paradigms. These range from discrete shifts modeled as unknown structural breaks (Corradi and Swanson 2014) and state-dependent transitions in threshold FAR models (Yan and Cheng 2022) to broad nonparametric specifications that allow for smooth parameter drift. The latter includes both functional varying-coefficient models (Li et al. 2020*b*) and time-varying coefficient regressions (Wei and Zhang 2020; Fu et al. 2024), reflecting the underlying assumption that economic relationships evolve as a continuous process rather than through episodic shocks. More recently, these ideas have been extended to time-varying forecast combinations and quantile regressions (Chen et al. 2024; Tu and Wang 2025).

Despite this progress, a key limitation remains the reliance on stationary factor structures; when nonstationarity is present, it is often eliminated through differencing, which discards essential low-frequency information (Choi and Jeong 2020). Although several studies develop estimation theory for nonstationary factors (Bai 2004; Eichler et al. 2011; Choi 2017), they do not integrate such factors into forecasting frameworks. Recent work by Hannadige et al. (2024) incorporates a mixture of stationary and nonstationary factors into a FAR model but imposes constant coefficients, leaving structural instability unaddressed.

Second, a parallel body of literature investigates nonlinear econometric relationships with nonstationary regressors, including cointegrating regressions with smoothly time-varying coefficients (Park and Hahn 1999; Li et al. 2020*a*), nonlinear functions of integrated time series (Park and Phillips 2001), semi-varying and additive models for mixed stationary and nonstationary regressors (Li et al. 2017; Dong and Linton 2018), and time-varying vector error-correction models (Gao et al. 2025). Despite these developments, the nonstationary regressors considered in this literature are typically assumed to be directly observable.

In many macro-financial forecasting problems, however, persistent drivers are not associated with any single observed series but instead manifest as common trends shared across a large panel

of indicators, such as the long-run components underlying price levels, monetary aggregates, and asset valuations. Extracting such latent persistent factors from high-dimensional data, rather than relying on a pre-selected observable proxy, is therefore essential for fully exploiting the low-frequency predictive content available in large macroeconomic datasets. In this paper, we address this gap by developing a forecasting framework that accommodates latent nonstationary factors and time-varying coefficients simultaneously, allowing for both static and dynamic (possibly cointegrated) nonstationary factor structures. This enables our model to capture structural instability while exploiting the low-frequency information embedded in persistent latent drivers.

The paper is organized as follows. Section 2 introduces the model setup and estimation approach. Section 3 presents assumptions and asymptotic results for the time-varying coefficient estimator with a static nonstationary factor structure, and Section 4 extends the model to a dynamic factor structure. Section 5 evaluates forecasting performance on U.S. inflation. Section 6 concludes. Monte Carlo evidence under both static and dynamic factor structures is reported in Appendix B of the online supplementary material. The supplementary file also contains all proofs and technical lemmas, together with a formal justification of the autoregressive reparameterization and additional details on the macroeconomic dataset.

Throughout the paper, I_k denotes a $k \times k$ identity matrix. For a real matrix A, let $\|A\|$ and $\|A\|_2$ denote the Frobenius and spectral norm, respectively. When A is symmetric, we use $\lambda_{\min}(A)$ and $\lambda_{\max}(A)$ to denote its minimum and maximum eigenvalues, respectively. For given matrices A and B, let $A \oplus B = \mathrm{diag}(A, B)$. For constants a, b, let $a \wedge b := \min(a, b)$ and $a \vee b := \max(a, b)$. $B(\cdot, \Sigma)$ denotes a Brownian motion with covariance matrix Σ, and $\lfloor \cdot \rfloor$ is the floor operator. We use $\Rightarrow$ to denote weak convergence, especially for stochastic processes and Brownian-functional limits, and use $\xrightarrow{p}$ and $\xrightarrow{d}$ to denote convergence in probability and in distribution, respectively.

2 Model Setup and Estimation

In this section, we propose a general factor-augmented forecasting regression that accommodates both time-varying coefficients and a factor structure:

$$y_{t+h} = \theta(\tau_t)' z_t + \varepsilon_{t+h}, \quad t = 1, ..., T - h, \tag{2.1}$$

where $\tau_t = t/T$, $\theta(\cdot)$ is an $r \times 1$ vector of deterministic smooth coefficient functions on $[0, 1]$ (we write $\theta_t = \theta(\tau_t)$ for notational convenience), and z_t is an $r \times 1$ generic predictor vector that may

contain observed predictors, Δy_t and its lags, and/or latent factors, with components that may be $I(0)$, $I(1)$, or a mixture of both. Setting the first element of z_t to one allows for the inclusion of an unknown deterministic trending term in the regression.

The latent factors accommodated by our framework follow the factor structure:

$$X_t = \Lambda F_t + e_t, \quad t = 1, \ldots, T, \tag{2.2}$$

where $X_t = (X_{1t}, \ldots, X_{Nt})'$ is an observable high-dimensional data vector, $\Lambda = (\lambda_1, \ldots, \lambda_N)'$ is the loading matrix, and $e_t = (e_{1t}, \ldots, e_{Nt})'$ is a vector of $I(0)$ idiosyncratic errors. We allow the r_F latent factors in F_t to be $I(0)$, $I(1)$, or a mixture of both, where the mixed case nests the dynamic factor model via its equivalent static representation; see Bai (2004) for details.

2.1 Examples

In what follows, we provide the reader with several examples to illustrate how the model (2.1) nests a broad range of old and new models as special cases. To this end, let Z_t be an $r_Z \times 1$ vector of observed $I(0)$ predictors, W_t an $r_W \times 1$ vector of observed $I(1)$ predictors, and F_t an $r_F \times 1$ vector of latent factors.

Example 1 (Pure observed $I(0)$ predictors)**.** Set $z_t = Z_t$. Then model (2.1) reduces to $y_{t+h} = \theta_Z(\tau_t)'Z_t + \varepsilon_{t+h}$, which recovers the time-varying regression model of Cai (2007).

Example 2 (Observed $I(0)$ predictors and $I(0)$ latent factors)**.** Set $z_t = (Z_t', F_t')'$ with F_t being $I(0)$. Then (2.1) reduces to $y_{t+h} = \theta_Z(\tau_t)'Z_t + \theta_F(\tau_t)'F_t + \varepsilon_{t+h}$. This case is directly connected to the time-varying diffusion index model of Wei and Zhang (2020) and the time-varying FAVAR framework of Fu et al. (2024).

Example 3 (Pure observed $I(1)$ predictors)**.** Set $z_t = W_t$. Then model (2.1) reduces to $y_{t+h} = \theta_W(\tau_t)'W_t + \varepsilon_{t+h}$, which recovers the time-varying cointegrating regression of Phillips et al. (2017).

Example 4 (Mixture of observed $I(0)$ and $I(1)$ predictors)**.** Set $z_t = (Z_t', W_t')'$. Then (2.1) reduces to $y_{t+h} = \theta_Z(\tau_t)'Z_t + \theta_W(\tau_t)'W_t + \varepsilon_{t+h}$, yielding a time-varying regression with mixed-persistence observed predictors.

Example 5 (**Mixed** $I(0)/I(1)$ **observable predictors and latent factors**)**.** Set $z_t = (Z_t', F_t', W_t')'$, where F_t exhibit mixed persistence. Then (2.1) reduces to

$$y_{t+h} = \theta_Z(\tau_t)'Z_t + \theta_F(\tau_t)'F_t + \theta_W(\tau_t)'W_t + \varepsilon_{t+h}.$$

This specification subsumes the predictor types in the preceding examples and serves as our aim.

Remark 1. A natural starting point for forecasting a persistent macro variable such as inflation is to combine its own lags, which capture nominal rigidities and expectation formation, with latent and observed persistent predictors. This leads to a Time-Varying Factor-Augmented Autoregressive Distributed Lag (TV-FA-ARDL) specification:

$$y_{t+h} = \theta_Y(\tau_t)'\bar{Y}_t + \theta_F(\tau_t)'F_t + \theta_W(\tau_t)'W_t + \theta_Z(\tau_t)'Z_t + \varepsilon_{t+h}, \tag{2.3}$$

where $\bar{Y}_t = (y_t, y_{t-1}, \ldots, y_{t-p+1})'$. While (2.3) is intuitive, it is not directly estimable in the present nonstationary environment. Since y_t is driven by the persistent predictors F_t and W_t, the level y_{t-i} and the vector $(F_t', W_t')'$ are exactly cointegrated. Consequently, the (appropriately normalized) sample moment matrix of the regressors in (2.3) is asymptotically singular. Note that each level lag y_{t-i} can nonetheless be decomposed into a telescoping sum of stationary increments Δy_{t-j+1} plus an $I(1)$ component. Collecting the increments into Z_t yields the equivalent TV-FAR specification in Example 5:

$$y_{t+h} = \theta_Z(\tau_t)'Z_t + \theta_F(\tau_t)'F_t + \theta_W(\tau_t)'W_t + \varepsilon_{t+h}, \tag{2.4}$$

which retains the predictive content of (2.3) while ensuring a non-degenerate moment matrix; a formal proof is given in an online supplementary document.

We emphasize that the present section focuses on the static nonstationary factor structure in (2.2), which rules out cointegration among the latent nonstationary factors. Our framework is then extended in Section 4 to include a more general dynamic factor model setting that allows F_t to contain both $I(0)$ and $I(1)$ components and accommodates cointegration. Having introduced the model framework, we next turn to estimation.

2.2 Estimation

When z_t includes latent factors generated from (2.2), the factor is first estimated by principal component analysis (PCA). Let $X = [X_1, \ldots, X_T]$ and $\widehat{F}$ denote the matrix of r_F eigenvectors of XX' corresponding to the largest r_F eigenvalues, normalized such that $\widehat{F}'\widehat{F}/T^2 = I_{r_F}$. The loading matrix is estimated by $\widehat{\Lambda} = X'\widehat{F}/T^2$. In practice, r_F can be selected using the Integrated Panel Criterion of Bai (2004). Since the factors are identified only up to rotation, under the normalization $\frac{F'F}{T^2} = I_{r_F}$ and diagonal $\Lambda'\Lambda$, $\widehat{F}$ recovers FH' for some nonsingular matrix $H = \widehat{V}_{NT}^{-1}(\widehat{F}'F/T^2)(\Lambda'\Lambda/N)$, where $\widehat{V}_{NT}$ is the diagonal matrix of the largest r_F eigenvalues of $\frac{XX'}{T^2N}$.

To unify the cases with and without latent factors, let $\widehat{z}_t$ denote the feasible regressor vector: if z_t contains latent factors, they are replaced by their PCA estimates; otherwise $\widehat{z}_t = z_t$. Correspondingly, define $\theta^*(\cdot)$ as the coefficient function associated with $\widehat{z}_t$, with $\theta^*(\cdot) = \theta(\cdot)$ when no latent factors are present. For the mixed specification $z_t = (Z_t', F_t', W_t')'$, we have $\widehat{z}_t = (Z_t', \widehat{F}_t', W_t')'$ and, conformably, $\theta^*(\tau) = (\theta_Z(\tau)', \theta_F^*(\tau)', \theta_W(\tau)')'$, where $\theta_F^*(\tau) = H^{-1\prime}\theta_F(\tau)$. The forecasting regression can then be written as

$$y_{t+h} = \theta^*(\tau_t)'\widehat{z}_t + \varepsilon_{t+h}^*, \qquad t = 1, \ldots, T-h, \tag{2.5}$$

where $\varepsilon_{t+h}^* = \varepsilon_{t+h} + \Delta_t$ and $\Delta_t := \theta_F^*(\tau_t)'(HF_t - \widehat{F}_t)$.

We then estimate the time-varying coefficient functions by an orthonormal series method. Here, we employ the sieve method instead of kernel estimation, because the resulting nonstationary kernel–weighted sample covariance matrix is asymptotically singular. In such case, one will need to deal with the so–called "kernel-induced degeneracy" in the unit-root literature (cf., Phillips et al. 2017) when a standard kernel method is employed.

Let $\Phi_k(\tau) = (\varphi_0(\tau), \ldots, \varphi_{k-1}(\tau))'$ be an orthonormal basis on $[0,1]$ (e.g., the cosine basis). Using a truncation parameter k, we write

$$\theta^*(\tau) = \Psi_k(\tau)'\beta + \gamma_k(\tau), \tag{2.6}$$

where $\Psi_k(\tau) = I_r \otimes \Phi_k(\tau)$, $\beta \in \mathbb{R}^{rk}$ stacks the series coefficients, and $\gamma_k(\tau)$ is the truncation residual. Define the regressor expansion $b_{t,k} = \Psi_k(\tau_t)\widehat{z}_t \in \mathbb{R}^{rk}$, and stack $B_{T,k} = (b_{1,k}, \ldots, b_{T-h,k})'$ and $Y = (y_{1+h}, \ldots, y_T)'$. The sieve estimator is then given by

$$\widehat{\beta} = (B_{T,k}'B_{T,k})^{-1}B_{T,k}'Y \quad \text{and} \quad \widehat{\theta^*}(\tau) = \Psi_k(\tau)'\widehat{\beta}. \tag{2.7}$$

3 Asymptotic Properties

In this section, we focus on the general specification

$$z_t = (Z_t', F_t', W_t')', \tag{3.1}$$

where Z_t is $I(0)$, W_t is $I(1)$, and F_t is an $I(1)$ latent factor vector following the static nonstationary factor model (2.2). The other examples in Section 2 can be obtained as special cases of (3.1) by dropping one or more of Z_t, F_t, and W_t; we briefly discuss how to establish the corresponding results for these simpler specifications.

3.1 Assumptions

To facilitate the asymptotic analysis, we make the following assumptions.

Assumption A.

(i) Define $x_t = (F_t', W_t')'$ as an $I(1)$ process such that $x_t = x_{t-1} + u_t$ with $u_t = (u_t^{F\prime}, u_t^{W\prime})'$. Let $\eta_t = (Z_t', u_t')'$ be a stationary causal process such that $\eta_t = G(\upsilon_t, \upsilon_{t-1}, \ldots)$, where υ_t are independent and identically distributed (i.i.d.) random vectors, $\Sigma_\eta = \sum_{k=-\infty}^{\infty} \mathbb{E}(\eta_0 \eta_k')$ is positive definite, and $G(\cdot) = [G_1(\cdot), \ldots, G_r(\cdot)]'$ is a vector of known measurable functions.

(ii) Let $|\eta_m - \eta_m^*|_4 = \lambda_4^f(m) = O\left(m^{-a}\right)$ for some $a > 2$, where $\eta_t^* = G(\upsilon_t, \ldots, \upsilon_1, \upsilon_0', \upsilon_{-1}, \ldots)$ is the coupled version of η_t and υ_0' is an independent copy of υ_0.

Assumption A imposes weak dependence conditions on $\{\eta_t\} = \{(Z_t', u_t')'\}$, allowing for serial correlation and conditional heteroscedasticity of unknown forms. The strength of temporal dependence is measured by the functional dependence measure of Wu (2005), which nests many stationary processes as special cases (see, e.g., Tong 1990, p. 204; Wu 2005). Intuitively, the physical dependence coefficients $\{\lambda_4^f(m)\}$ quantify the dependence of the output η_m on υ_0. We only require an algebraic decay rate $a > 2$, which is sufficient to ensure the applicability of the Gaussian approximation established in Karmakar and Wu (2020). Specifically, on a possibly richer probability space, one can construct a Brownian motion $B(r, \Sigma_u)$ such that

$$\sup_{0 \le r \le 1} \left\| \frac{1}{\sqrt{T}} \sum_{t=1}^{\lfloor Tr \rfloor} u_t - B(r, \Sigma_u) \right\| = O_P\left(T^{-1/4}\right), \tag{3.2}$$

where Σ_u denotes the long-run covariance matrix of u_t. Note that this Gaussian approximation is commonly used in the literature on nonlinear and nonstationary regression models (e.g., Park and Hahn 1999; Cai et al. 2009; Chen et al. 2015). We use it to derive the consistency of nonstationary series-weighted sample covariances together with their convergence rates.

The assumption that Σ_η is positive definite rules out cointegration among the components of F_t. Cointegration is equivalent to the presence of both $I(1)$ and $I(0)$ common factors, which can be accommodated by a dynamic nonstationary factor model considered in Section 4 below.

Assumption B. Suppose that $c_1 \le \lambda_{\min}(\Sigma_\Lambda) \le \lambda_{\max}(\Sigma_\Lambda) \le c_2$ and $\max_{i \in [N]} \|\lambda_i\| \le c_0$ for some constants $c_0, c_1, c_2 > 0$, where $\Sigma_\Lambda := \lim_{N \to \infty} \frac{1}{N} \Lambda' \Lambda$.

Assumption C.

(i) Let e_t be a stationary process such that $e_t = g(\upsilon_t, \upsilon_{t-1}, \ldots)$ with $\mathbb{E}(e_t) = 0_N$, $\mathbb{E}(e_t e_t') = \Sigma_e$, $\|\Sigma_e\|_2 \leq M$, and $g(\cdot) = [g_1(\cdot), \ldots, g_N(\cdot)]'$ being a vector of known measurable functions.

(ii) Let $\sup_{q \in \mathbb{R}^N, \|q\|<\infty} \left|\bar{e}_{m,q} - \bar{e}^*_{m,q}\right|_4 = \lambda^e_4(m) = O\left(m^{-a}\right)$ for some $a > 2$, where $\bar{e}_{t,q} = e_t' q$ and $\bar{e}^*_{t,q} = e^{*\prime}_t q$ with $e^*_t = g\left(\upsilon_t, \ldots, \upsilon_1, \upsilon_0', \upsilon_{-1}, \ldots\right)$, and υ_0' is as defined in Assumption A(i).

(iii) $\left|\mathcal{P}_{t-m}\left(\frac{1}{\sqrt{N}}\sum_{i=1}^N e_{it}e_{is}\right)\right|_2 = O\left(m^{-b}\right)$ for some $b > 1$, where $\mathcal{P}_t[\cdot] = \mathbb{E}\left[\cdot \mid \mathscr{E}_t\right] - \mathbb{E}\left[\cdot \mid \mathscr{E}_{t-1}\right]$ and $\mathscr{E}_t = (\upsilon_t, \upsilon_{t-1}, \ldots)$.

Assumption B treats the loadings as non-random and ensures that each factor makes a non-trivial contribution to the variance of X_t; see, e.g., Bai (2003) and Su and Wang (2017). The asymptotic analysis remains valid when the loadings are random and independent of the factors and error terms. Assumption C allows for limited dependence in the idiosyncratic component across both the time-series and cross-sectional dimensions, thereby characterizing an approximate factor model that generalizes strict factor models. Specifically, this assumption accommodates various unknown forms of cross-sectional dependence by regulating the decay rate of temporal dependence after taking weighted cross-sectional averages. These conditions on $\{e_t\}$ can be readily verified for a general class of high-dimensional time series, such as high-dimensional MA(∞) and GARCH processes (e.g., Gao et al. 2024*c*).

Assumption D. Let $\Sigma_F = \int_0^1 B_F(r)B_F(r)'\mathrm{d}r$, where $B_F(r)$ is a Brownian motion vector with covariance matrix Σ_{u^F} (the submatrix of Σ_η associated with u^F_t). We impose the following conditions: (i) The eigenvalues of $(\Sigma_\Lambda \cdot \Sigma_F)$ are distinct; (ii) $\left|\mathcal{P}_{t-m}\left(\frac{1}{T}\frac{1}{\sqrt{N}}F_t\sum_{i=1}^N e_{it}e_{is}\right)\right|_2 = O\left(m^{-b}\right)$ for some $b > 1$.

Assumption D(i) guarantees a unique limit for $\widehat{F}'F/T^2$, which appears in the limiting distributions of $\widehat{F}_t$. Assumption D(ii) implies the underlying data generating process to satisfy the following moment restriction (see Lemma A.1(iv) of the online supplement for details):

$$\left|\frac{1}{T}\frac{1}{\sqrt{N}}\sum_{t=1}^T F_t \sum_{i=1}^N \left[e_{it}e_{is} - \mathbb{E}\left(e_{it}e_{is}\right)\right]\right|_2 = O(1).$$

This is similar to the widely adopted conditions in the literature on factor models (see, e.g., Assumption F.1 in Bai 2003, Assumption A.1(vii) in Su and Wang 2017 and Assumption A.7 in Fu et al. 2023) and is not stringent because the sums in Assumption D involve zero mean random variables.

Assumption E. Let $\theta(\cdot) = (\theta_1(\cdot), \ldots, \theta_r(\cdot))'$, and define $s = \min_{1\leq i\leq r} s_i$, where $\theta_i(\cdot)$ is continuously differentiable of order s_i, as $T, N \to \infty$, (i) $T^{-1/4}k = o(1)$; (ii) $Tk^{1/2-s} = o(1)$; (iii) $(T/N)^{1/2}k = o(1)$.

Assumption E requires the truncation parameter k to diverge at appropriate rate. Assumption E(i) ensures the consistency of nonstationary series-weighted sample covariances in Lemma 3.1. The condition involves the term $T^{-1/4}$ arising from the Gaussian approximation (3.2). The smoothness conditions in Assumption E(ii) ensure that the truncation error of the series approximation is negligible and enable us to establish a desirable rate of convergence. Assumption E(iii) guarantees that the estimation error arising from the use of estimated factors remains asymptotically insignificant. Compared with constant-coefficient FAR model, the interaction between the basis functions and $(HF_t - \widehat{F}_t)$ amplifies the factor estimation error. As a result, more stringent conditions on both T and N are required.

We stress that these requirements on k are mutually compatible. For instance, if we choose $k = \lfloor T^\tau \rfloor$ with $0 < \tau < \frac{1}{4}$ and $s > \frac{1}{\tau} + \frac{1}{2}$, and impose $\frac{T^{1+2\tau}}{N} \to 0$, then Assumption E is satisfied.

Assumption F. Let $\{\varepsilon_t\}$ be independent of $\{\eta_t\}$. For any $h > 0$, $\mathbb{E}(\varepsilon_{t+h} \mid \mathcal{B}_{NT,t}) = 0$ and $\mathbb{E}(\varepsilon_{t+h}^2 \mid \mathcal{B}_{NT,t}) = \sigma_\varepsilon^2$ almost surely (a.s.), where $\mathcal{B}_{NT,t} = \sigma(\varepsilon_t, \varepsilon_{t-1}, \ldots)$. Moreover, $\max_{t\leq T} \mathbb{E}(|\varepsilon_{t+h}|^{4+\kappa} \mid \mathcal{B}_{NT,t}) < \infty$ a.s. for some $\kappa > 0$.

Assumption F is appropriate in the context of forecasting. We assume that $\{\varepsilon_t\}$ is independent of other random quantities, which is standard in the factor-augmented regression (Bai and Ng 2006).

3.2 Asymptotic Results

Before presenting the asymptotic properties of the estimators, we introduce some notation and preliminary results. Let U_k be a 3×3 symmetric block matrix with elements defined as

$$
\begin{aligned}
U_{k,11} &= \mathbb{E}[Z_t Z_t'] \otimes I_k, \quad U_{k,12} = \int_0^1 \mathbb{E}(Z_t) B_F(r)' \otimes \Phi_k(r)\Phi_k(r)' \mathrm{d}r \, (H' \otimes I_k), \\
U_{k,13} &= \int_0^1 \mathbb{E}(Z_t) B_W(r)' \otimes \Phi_k(r)\Phi_k(r)' \mathrm{d}r, \\
U_{k,22} &= (H \otimes I_k) \int_0^1 B_F(r) B_F(r)' \otimes \Phi_k(r)\Phi_k(r)' \mathrm{d}r \, (H' \otimes I_k), \\
U_{k,23} &= (H \otimes I_k) \int_0^1 B_F(r) B_W(r)' \otimes \Phi_k(r)\Phi_k(r)' \mathrm{d}r,
\end{aligned}
$$

$$U_{k,33} = \int_0^1 B_W(r)B_W(r)' \otimes \Phi_k(r)\Phi_k(r)'\mathrm{d}r,$$

where $B_F(r) = B(r, \Sigma_{u^F})$ and $B_W(r) = B(r, \Sigma_{u^W})$ are multivariate Brownian motions with covariance matrices Σ_{u^F} (defined in Assumption A, conformably) and $\Sigma_{u^W} = \sum_{k=-\infty}^{\infty} \mathbb{E}\left(u_0^W u_k^{W\prime}\right)$, respectively.

Lemma 3.1. *Under Assumptions A–E(i), as $N, T \to \infty$, we have $\|S_{1T}^{-1} B_{T,k}^{0\prime} B_{T,k}^0 S_{1T}^{-1} - U_k\| = o_P(1)$ on a richer probability space, where $S_{1T} = \mathrm{diag}(\sqrt{T} I_{kr_Z}, T I_{k(r_F+r_W)})$, $B_{T,k}^0 = (b_{1,k}^0, \ldots, b_{T-h,k}^0)'$ and $b_{t,k}^0 = \Psi_k(\tau_t) z_t^*$ with $z_t^* = (Z_t', (HF_t)', W_t')'$.*

Lemma 3.1 shows that $S_{1T}^{-1} B_{T,k}^{0\prime} B_{T,k}^0 S_{1T}^{-1}$ converges in probability to the stochastic limit U_k. Consequently, if the eigenvalues of U_k are bounded away from zero and infinity, then $S_{1T}^{-1} B_{T,k}^{0\prime} B_{T,k}^0 S_{1T}^{-1}$ is asymptotically nonsingular. With this preliminary result in hand, we now establish the asymptotic normality of the sieve estimator $\widehat{\theta}^*(\cdot)$.

Theorem 3.2. *Suppose that the eigenvalues of U_k are uniformly bounded away from zero and infinity. Let Assumptions A–F hold and $D_{1T} = \mathrm{diag}(\sqrt{T} I_{r_Z}, T I_{r_F+r_W})$. Then for any $\tau \in [0,1]$, as $N, T \to \infty$,*

$$\Sigma_{\theta,T}^{-1/2}(\tau) \underline{\Psi}_k^{-1}(\tau) D_{1T} \left(\widehat{\theta}^*(\tau) - \theta^*(\tau)\right) \xrightarrow{d} N(0, I_r),$$

where $\Sigma_{\theta,T}(\tau) = \sigma_\varepsilon^2 \overline{\Psi}_k(\tau)' (S_{1T}^{-1} B_{T,k}^{0\prime} B_{T,k}^0 S_{1T}^{-1})^{-1} \overline{\Psi}_k(\tau)$, $\overline{\Psi}_k(\tau) = \Psi_k(\tau)\underline{\Psi}_k^{-1}(\tau)$, and $\underline{\Psi}_k(\tau) = \|\Phi_k(\tau)\| I_r$.

Remark 2. (i) Theorem 3.2 establishes the asymptotic normality of the sieve estimator $\widehat{\theta}^*(\tau)$. The normalization $\underline{\Psi}_k^{-1}(\tau) D_{1T}$ reveals component-specific convergence rates: the estimation errors are of order $\|\Phi_k(\tau)\|/\sqrt{T}$ for the stationary components associated with Z_t, and of order $\|\Phi_k(\tau)\|/T$ for the nonstationary components associated with F_t and W_t. The former coincides with the rate of the sieve estimator for stationary components (Dong et al. 2023), while the faster rate for the integrated regressors reflects their stronger signal, in line with the literature on mixed-persistence regressors (Dong and Linton 2018).

(ii) The independence between the model errors $\{\varepsilon_t\}$ and the regressors, imposed in Assumption F, leads to a standard normal limit after normalization; see Theorem 2.3 of Phillips and Park (1988) for related discussions. Lemma 3.1, together with the eigenvalue condition on U_k, guarantees that the normalization matrix $\Sigma_{\theta,T}(\tau)$ is nonsingular with probability approaching one. This result offers the theoretical justification for the construction of confidence intervals. In practice, σ_ε^2 can

be consistently estimated by $\widehat{\sigma}_\varepsilon^2 = (T-h)^{-1}\sum_{t=1}^{T-h}\widehat{\varepsilon}_{t+h}^2$ with $\widehat{\varepsilon}_{t+h} = y_{t+h} - \widehat{\theta}_t^{*\prime}\widehat{z}_t$. Then, Lemma C.8 in the online supplement implies that replacing the oracle sample moment matrix and the innovation variance by their feasible counterparts yields a consistent estimator of $\Sigma_{\theta,T}(\tau)$:

$$\widehat{\Sigma}_{\theta,T}(\tau) = \widehat{\sigma}_\varepsilon^2 \overline{\Psi}_k(\tau)' \left(S_{1T}^{-1} B_{T,k}' B_{T,k} S_{1T}^{-1}\right)^{-1} \overline{\Psi}_k(\tau).$$

(iii) The limit is centered at the rotated coefficient $\theta^*(\tau)$, which absorbs the rotation indeterminacy from factor estimation. No bias term appears: under the rate restrictions in Assumption E, both the sieve approximation error and the factor-estimation error are asymptotically negligible, so the limit is driven entirely by the innovation ε_{t+h}.

3.3 Prediction Intervals

We next discuss the theoretical properties of the estimated conditional mean of (2.1), which form the basis for constructing confidence intervals for the conditional mean $y_{T+h|T}$ and prediction intervals for the forecasted variable y_{T+h}.

Define the feasible prediction at time T as $\widehat{y}_{T+h|T} = \widehat{\theta}^*(\tau_{T-h})'\widehat{z}_T$. The decomposition

$$\widehat{y}_{T+h|T} - y_{T+h|T} = (\widehat{\theta}^*(\tau_{T-h}) - \theta^*(\tau_{T-h}))'\widehat{z}_T + \theta_F(\tau_{T-h})' H^{-1}(\widehat{F}_T - HF_T)$$

has two components, arising from the estimation of $\theta^*(\cdot)$ and F_t, respectively.

The first term is studied in Theorem 3.2. To characterize the second term, we additionally impose a cross-sectional central limit condition on the loading-weighted idiosyncratic errors.

Assumption G. For each t, as $N \to \infty$,

$$\frac{1}{\sqrt{N}}\sum_{i=1}^{N} \lambda_i e_{it} \xrightarrow{d} N\left(0, \Gamma_t\right),$$

where $\Gamma_t = \lim_{N\to\infty} \frac{1}{N}\sum_{i=1}^{N}\sum_{j=1}^{N} \mathbb{E}\left(\lambda_i \lambda_j' e_{it} e_{jt}\right)$.

Assumptions B and C already imply $N^{-1/2}\sum_{i=1}^{N}\lambda_i e_{it} = O_P(1)$; Assumption G strengthens this stochastic boundedness to a normal limiting distribution, paralleling Assumption G of Bai (2004). Under Assumptions A–C and G, the result of Bai (2004) applies: if $N/T^3 \to 0$, then for each t,

$$\sqrt{N}\left(\widehat{F}_t - HF_t\right) \xrightarrow{d} N(0, V^{-1} Q \Gamma_t Q' V^{-1}) = N(0, \mathrm{Avar}(\widehat{F}_t)),$$

where $Q = V^{1/2}\Upsilon'\Sigma_\Lambda^{-1/2}$, $V = \mathrm{diag}\left(v_1, v_2, \ldots, v_{r_F}\right)$ contains the r_F largest eigenvalues of $\Sigma_\Lambda \Sigma_F$, Υ is the corresponding eigenvector matrix with $\Upsilon'\Upsilon = I_{r_F}$, and Γ_t is defined in Assumption G.

Combining these two results yields the asymptotic distribution of $\widehat{y}_{T+h|T}$.

Theorem 3.3. *Let* $\widehat{y}_{T+h|T} = \widehat{\theta}^{*\prime}_{T-h}\widehat{z}_T$. *Under Assumptions A–G and the condition* $N/T^3 \to 0$,

$$\frac{\widehat{y}_{T+h|T} - y_{T+h|T}}{\sqrt{\mathrm{var}(\widehat{y}_{T+h|T})}} \xrightarrow{d} N(0,1),$$

where $\mathrm{var}(\widehat{y}_{T+h|T}) = \widehat{z}_T' D_{1T}^{-1} \underline{\Psi}_k(\tau_{T-h})' \Sigma_{\theta,T}(\tau_{T-h}) \underline{\Psi}_k(\tau_{T-h}) D_{1T}^{-1} \widehat{z}_T + \frac{1}{N} \widehat{\theta}^{*\prime}_F(\tau_{T-h}) \,\mathrm{Avar}(\widehat{F}_T) \widehat{\theta}^{*}_F(\tau_{T-h})$.

The forecast error variance has two sources that decay at different rates: a coefficient-estimation component of order $O_P(k/T)$ and a factor-estimation component of order $O_P(1/N)$. The overall convergence rate of $\widehat{y}_{T+h|T}$ is therefore $\sqrt{N} \wedge \sqrt{T/k}$, with the binding source determined by the relative magnitude of N and T/k. Ignoring the factor-estimation component would yield overly narrow confidence intervals for the conditional mean, particularly when N is moderate.

When the target is the realized value y_{T+h} rather than its conditional mean, the forecast error decomposes as

$$\widehat{y}_{T+h|T} - y_{T+h} = \left(\widehat{y}_{T+h|T} - y_{T+h|T}\right) - \varepsilon_{T+h}.$$

If $\varepsilon_{T+h} \sim N(0, \sigma_\varepsilon^2)$, the forecast error is asymptotically normal with variance

$$\mathrm{var}(\widehat{\varepsilon}_{T+h}) = \mathrm{var}(\widehat{y}_{T+h|T} - y_{T+h}) = \sigma_\varepsilon^2 + \mathrm{var}(\widehat{y}_{T+h|T}).$$

Valid prediction intervals can be obtained by replacing σ_ε^2, $\Sigma_{\theta,T}(\tau_{T-h})$, and $\mathrm{Avar}(\widehat{F}_T)$ with their consistent estimates in practice. Specifically, $\mathrm{Avar}(\widehat{F}_T)$ can be estimated by using the method of Bai and Ng (2006).

4 Extension to Dynamic Factor Structure

Consider the following dynamic nonstationary factor model:

$$X_{it} = \gamma_i(L)' E_t + e_{it} \quad \text{with} \quad E_t = E_{t-1} + u_t^E, \tag{4.1}$$

where $\gamma_i(L)$ is a vector of lag polynomials, E_t is an $r_E \times 1$ vector of $I(1)$ factors, and u_t^E is an $r_E \times 1$ vector of innovations. The relationship between X_{it} and E_t is dynamic here, whereas in (2.1) it is static, even though the factors themselves may exhibit arbitrary dynamics. Following the argument in Bai (2004), even if $\gamma_i(L)$ is an infinite-order lag polynomial, i.e., $\gamma_i(L) = \sum_{j=0}^{\infty} \gamma_{ij} L^j$ with $\sum_{j=0}^{\infty} j\,|\gamma_{ij}| < \infty$, the dimension of the common factors can still be consistently estimated. However, to derive the distributional theory for the model, we restrict $\gamma_i(L)$ to be a finite-order polynomial $\gamma_i(L) = \sum_{j=0}^{s} \gamma_{ij} L^j$.

Under this restriction, the dynamic nonstationary factor model can be rewritten as a static

nonstationary factor model involving both $I(1)$ and $I(0)$ factors. Specifically,

$$\begin{aligned} X_{it} &= \gamma_{i0}' E_t + \gamma_{i1}' E_{t-1} + \cdots + \gamma_{is}' E_{t-s} + e_{it} \\ &= \lambda_{i0}' E_t - \lambda_{i1}' \Delta E_t - \cdots - \lambda_{is}' \Delta E_{t-s+1} + e_{it}, \end{aligned} \tag{4.2}$$

where $\lambda_{ik} = \gamma_{ik} + \gamma_{i,k+1} + \cdots + \gamma_{is}$. Define $\lambda_i' = (\lambda_{i0}', -\lambda_{i1}', \ldots, -\lambda_{is}')$ and $F_t = \left(E_t', \Delta E_t', \ldots, \Delta E_{t-s+1}'\right)'$. Then (4.2) can be rewritten as

$$X_{it} = \lambda_i' F_t + e_{it} = \lambda_{i0}' E_t + \lambda_{i,0-}' G_t + e_{it}, \tag{4.3}$$

where $G_t = \left(\Delta E_t', \ldots, \Delta E_{t-s+1}'\right)'$ and $\lambda_{i,0-}$ denotes the subvector of λ_i excluding λ_{i0}. This reparameterization shows that a dynamic nonstationary factor model can be expressed as a static factor model containing both $I(1)$ and $I(0)$ common factors.

The dynamic specification offers greater flexibility than its static counterpart, with two main theoretical advantages. First, it allows for general $I(0)$ factors: the process G_t need not consist solely of lags of ΔE_t, as the key requirement for our analysis is only that G_t is $I(0)$. Second, it naturally accommodates cointegration among the $I(1)$ factors. Suppose E_t is a vector of cointegrated $I(1)$ factors. Then there exists an invertible matrix P such that $PE_t = (\xi_t', \psi_t')'$, where ξ_t is a vector of non-cointegrated $I(1)$ factors and ψ_t is a vector of $I(0)$ factors formed as linear combinations of the components of E_t. With a slight abuse of notation, we let $\xi_t = E_t$, $\psi_t = G_t$, and redefine $\lambda_i' P^{-1}$ as λ_i', so that cointegrated $I(1)$ factors can still be expressed in the form of (4.3).

In light of the above discussion, the dynamic nonstationary factor model (4.1) admits an equivalent static representation involving both non-cointegrated $I(1)$ and general $I(0)$ factors, with the same asymptotic properties:

$$X_{it} = \lambda_i' F_t + e_{it} = \lambda_i^{E\prime} E_t + \lambda_i^{G\prime} G_t + e_{it}, \tag{4.4}$$

where $\lambda_i = (\lambda_i^{E\prime}, \lambda_i^{G\prime})'$, and $F_t = (E_t', G_t')'$ is an $r_F \times 1$ vector that mixes nonstationary and stationary factors, with E_t an $r_E \times 1$ vector of $I(1)$ factors and G_t an $r_G \times 1$ vector of $I(0)$ factors.

The representation in (4.4) allows us to incorporate a dynamic nonstationary factor structure into the forecasting regression (2.1) by working with an equivalent static factor vector containing both $I(1)$ and $I(0)$ components. This yields a particularly general specification that combines stationary and nonstationary predictors and a mixture of nonstationary and stationary latent factors, which serves as the main setting considered in this section. It also nests the static specification in Section 3 as a special case when the dynamic factor structure reduces to a purely nonstationary

static factor representation. The following example summarizes the resulting formulation.

Example 6 (**Mixed observed $I(0)/I(1)$ predictors and mixed $I(0)/I(1)$ latent factors**). Let $z_t = (Z_t', F_t', W_t')'$, where Z_t is $I(0)$, W_t is $I(1)$, and F_t is generated from the dynamic nonstationary factor model (4.1). Using the static-equivalent representation (4.4), we partition the latent factors as $F_t = (E_t', G_t')'$, where E_t collects the $I(1)$ latent trends and G_t collects the $I(0)$ latent components, so that $z_t = (Z_t', E_t', G_t', W_t')'$. The corresponding specification,

$$y_{t+h} = \theta_Z(\tau_t)'Z_t + \theta_E(\tau_t)'E_t + \theta_G(\tau_t)'G_t + \theta_W(\tau_t)'W_t + \varepsilon_{t+h},$$

extends (2.1) by allowing a mixture of nonstationary and stationary latent factors.

4.1 Estimation

Under the static-equivalent representation (4.4), let $F_t = (E_t', G_t')'$, where E_t collects r_E $I(1)$ factors and G_t collects r_G $I(0)$ factors, with $r_F = r_E + r_G$. The dimensions (r_E, r_G) can be consistently selected by the Integrated Panel Criterion of Bai (2004). Let $\widetilde{E}$ denote T times the first r_E eigenvectors of XX', normalized such that $\widetilde{E}'\widetilde{E}/T^2 = I_{r_E}$, and let $\widetilde{G}$ denote $\sqrt{T}$ times the next r_G eigenvectors, normalized such that $\widetilde{G}'\widetilde{G}/T = I_{r_G}$. The estimated factors are collected as $\widetilde{F} = (\widetilde{E}, \widetilde{G})$, and the loading estimator is $\widetilde{\Lambda} = X'\widetilde{F}D_T^{-2}$, where $D_T = \mathrm{diag}(TI_{r_E}, \sqrt{T}I_{r_G})$. Let $\widetilde{V}_{NT} = \mathrm{diag}(\widetilde{V}_{NT}^E, \widetilde{V}_{NT}^G)$ collect the first r_E eigenvalues of $XX'/(T^2N)$ and the next r_G eigenvalues of $XX'/(TN)$, respectively. Under the identification restrictions $D_T^{-2}F'F = I_{r_F}$ and diagonal $\Lambda'\Lambda$, $\widetilde{F}$ recovers $F\underline{H}'$, where $\underline{H} = \widetilde{V}_{NT}^{-1}(D_T^{-2}\widetilde{F}'F)(\Lambda'\Lambda/N)$.

Given $\widetilde{F}_t = (\widetilde{E}_t', \widetilde{G}_t')'$, the feasible predictor is $\widetilde{z}_t = (Z_t', \widetilde{E}_t', \widetilde{G}_t', W_t')'$. The sieve estimation in (2.7) is then carried out with $\widehat{z}_t$ replaced by $\widetilde{z}_t$, yielding the estimator

$$\widetilde{\theta}^*(\tau) = \big(\widetilde{\theta}_Z'(\tau),\ \widetilde{\theta}_E^{*\prime}(\tau),\ \widetilde{\theta}_G^{*\prime}(\tau),\ \widetilde{\theta}_W'(\tau)\big)'$$

for the coefficient function: $\theta^*(\tau) = \big(\theta_Z'(\tau),\ \theta_E^{*\prime}(\tau),\ \theta_G^{*\prime}(\tau),\ \theta_W'(\tau)\big)'$, where $(\theta_E^{*\prime}(\tau), \theta_G^{*\prime}(\tau))' = \underline{H}^{-1\prime}\theta_F(\tau)$ is partitioned conformably with F_t.

4.2 Asymptotic Properties

Because of the dynamic factor structure, Assumptions A and D are replaced by the following corresponding conditions.

Assumption A*.

(i) Define $x_t = (E_t', W_t')'$ as an $I(1)$ process such that $x_t = x_{t-1} + u_t$ with $u_t = (u_t^{E\prime}, u_t^{W\prime})'$. Let $\eta_t = (Z_t', G_t', u_t')'$ be a stationary causal process such that $\eta_t = G(\upsilon_t, \upsilon_{t-1}, \ldots)$ with υ_t being i.i.d. random vectors, $\Sigma_\eta = \sum_{k=-\infty}^{\infty} \mathbb{E}(\eta_0 \eta_k')$ is positive definite, and $G(\cdot) = [G_1(\cdot), \ldots, G_r(\cdot)]'$ is a vector of known measurable functions.

(ii) Let $|\eta_m - \eta_m^*|_4 = \lambda_4^f(m) = O\left(m^{-a}\right)$ for some $a > 2$, where $\eta_t^* = G(\upsilon_t, \ldots, \upsilon_1, \upsilon_0', \upsilon_{-1}, \ldots)$ is the coupled version of η_t and υ_0' is an independent copy of υ_0;

(iii) For the idiosyncratic error e_t, $\mathbb{E}(e_t \mid G_t) = 0_N$ a.s. for all t.

Assumption A* is analogous to Assumption A, except that it additionally quantifies the temporal dependence of the stationary factors G_t. Condition (iii) is a standard orthogonality condition for stationary factor components. It is used to derive the factor-estimation error bound under a mixed-persistence factor structure, while still allowing for higher-order dependence between G_t and e_t and the cross-sectional dependence in e_t permitted by Assumption C. Similar or stronger exogeneity conditions are commonly imposed in related factor-model studies; see, for example, Su and Wang (2017); Fu et al. (2024); Peng et al. (2025); Su et al. (2025).

Assumption D*. Let $D_T = \mathrm{diag}(TI_{r_E}, \sqrt{T} I_{r_G})$ and $\Sigma_F = \int_0^1 B_E(r) B_E(r)' \mathrm{d}r \oplus \Sigma_G$, where $B_E(r)$ is a vector Brownian motion with covariance matrix Σ_{u^E} (the submatrix of Σ_η associated with u_t^E), and Σ_G is the probability limit of $T^{-1} \sum_{t=1}^T G_t G_t'$. We impose the following conditions:

(i) the eigenvalues of $\Sigma_\Lambda \cdot \Sigma_F$ are distinct;

(ii) $\left| \mathcal{P}_{t-m} \left(N^{-1/2} D_T^{-1} F_t \sum_{i=1}^N e_{it} e_{is} \right) \right|_2 = O\left(m^{-b}\right)$ for some $b > 1$.

Assumption D* is analogous to Assumption D, but differs in the normalizing matrix D_T, which reflects the distinct time-series features of E_t and G_t.

The following theorem extends Lemma 3.1 and Theorems 3.2–3.3 to the dynamic factor specification.

Theorem 4.1.

(i) Under Assumptions A, B, C, D*, and E(i), there exists a symmetric block matrix V_k such that $\|S_{2T}^{-1} B_{T,k}^{0\prime} B_{T,k}^0 S_{2T}^{-1} - V_k\| = o_P(1)$ on a richer probability space, where $S_{2T} = \mathrm{diag}(\sqrt{T} I_{kr_Z}, TI_{kr_E}, \sqrt{T} I_{kr_G}, TI_{kr_W})$, $B_{T,k}^0 = (b_{1,k}^0, \ldots, b_{T-h,k}^0)'$ and $b_{t,k}^0 = \Psi_k(\tau_t) z_t^*$ with $z_t^* = (Z_t', (\underline{H} F_t)', W_t')'$.*

(ii) Suppose further that the eigenvalues of V_k are uniformly bounded away from zero and infinity, and that Assumptions A, B, C, D*, E and F hold. Let $D_{2T} = \text{diag}(\sqrt{T}I_{r_Z}, D_T, TI_{r_W})$. Then, for any $\tau \in [0,1]$, as $N, T \to \infty$,*

$$\Omega_{\theta,T}^{-1/2}(\tau)\underline{\Psi}_k^{-1}(\tau)D_{2T}(\widetilde{\theta}^*(\tau) - \theta^*(\tau)) \xrightarrow{d} N(0, I_r), \tag{4.5}$$

where $\Omega_{\theta,T}(\tau) = \sigma_\varepsilon^2 \overline{\Psi}_k(\tau)'(S_{2T}^{-1}B_{T,k}^{0\prime}B_{T,k}^0 S_{2T}^{-1})^{-1}\overline{\Psi}_k(\tau)$.

(iii) Let $\widetilde{y}_{T+h|T} = \widetilde{\theta}_{T-h}^{*\prime}\widetilde{z}_T$. Under Assumptions A, B, C, D*, E–G, together with the condition $N/T^2 \to 0$,*

$$\frac{\widetilde{y}_{T+h|T} - y_{T+h|T}}{\sqrt{\text{var}(\widetilde{y}_{T+h|T})}} \xrightarrow{d} N(0,1),$$

where

$$\text{var}(\widetilde{y}_{T+h|T}) = \widetilde{z}_T' D_{2T}^{-1}\underline{\Psi}_k(\tau_{T-h})'\Omega_{\theta,T}(\tau_{T-h})\underline{\Psi}_k(\tau_{T-h})D_{2T}^{-1}\widetilde{z}_T + \frac{1}{N}\widetilde{\theta}_F^{*\prime}(\tau_{T-h})\,\text{Avar}(\widetilde{F}_T)\widetilde{\theta}_F^*(\tau_{T-h}).$$

If we further assume $\varepsilon_t \sim N(0, \sigma_\varepsilon^2)$, then $\widetilde{\varepsilon}_{T+h} \sim N(0, \sigma_\varepsilon^2 + \text{var}(\widetilde{y}_{T+h|T}))$.

Remark 3. (i) Theorem 4.1 extends Theorem 3.2 to the dynamic nonstationary factor structure of Bai (2004), and generalizes the mixture-FAR framework of Hannadige et al. (2024) by allowing the forecasting coefficients to vary smoothly over time. The latent factor block is now partitioned into an integrated trend block E_t and a stationary cyclical block G_t, so the normalization $D_{2T} = \text{diag}(\sqrt{T}I_{r_Z}, TI_{r_E}, \sqrt{T}I_{r_G}, TI_{r_W})$ assigns the appropriate rates to stationary and integrated components, allowing mixed-rate inference to apply symmetrically to observed and latent regressors. Similarly to Remark 2(ii), by Lemma D.7 in the supplement, a consistent estimator of $\Omega_{\theta,T}(\tau)$ is given by $\widetilde{\Omega}_{\theta,T}(\tau) = \widehat{\sigma}_\varepsilon^2 \overline{\Psi}_k(\tau)' \left(S_{2T}^{-1}B_{T,k}'B_{T,k}S_{2T}^{-1}\right)^{-1}\overline{\Psi}_k(\tau)$.

(ii) The stronger condition $N/T^2 \to 0$ in part (iii), relative to $N/T^3 \to 0$ in Theorem 3.3, originates from the remainder term in the factor-estimation expansion. In the purely nonstationary static case, the leading remainder is governed by an integrated-factor term, whose negligibility after $\sqrt{N}$-normalization requires $N/T^3 \to 0$. Under the dynamic mixed $I(1)/I(0)$ representation, the stationary-factor block introduces an additional remainder term that decays at a slower rate, yielding the tighter requirement $N/T^2 \to 0$.

5 An Empirical Application

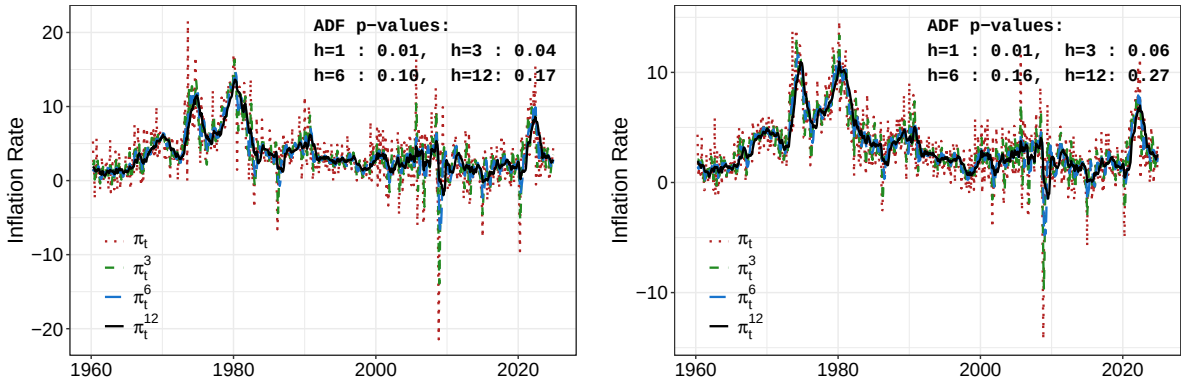
Forecasting the evolution of the general price level over long horizons is central to private-sector decision-making, given the prevalence of long-term nominal commitments. This task is challenging

because inflation is driven by a high-dimensional set of macroeconomic and financial variables, and the associated predictive relationships may vary over time due to changes in monetary policy, business-cycle conditions, and external shocks. To address these issues, this section applies our proposed time-varying factor-augmented forecasting regression model to predict U.S. inflation and evaluates its performance against several benchmark models.

5.1 Data and Forecasting Models

5.1.1 Data

We focus on two headline measures of consumer prices: the Consumer Price Index for All Urban Consumers (CPI) and the Personal Consumption Expenditures (PCE) deflator. The CPI is a widely used measure of inflation, while the PCE deflator is central to Federal Reserve policy analysis. Both series are seasonally adjusted and obtained from the Federal Reserve Economic Data (FRED) database.



(a) CPI (b) PCE

Figure 5.1: Inflation rates based on the two price indices at different horizons. The corresponding ADF test p-values are reported within the plots.

Following Stock and Watson (1999, 2002*a*), we model the CPI and PCE price indices as $I(2)$ processes in logarithms and focus on forecasting average monthly inflation. Let monthly inflation (at an annual rate) be $\pi_t = 1200 \ln(P_t/P_{t-1})$, where P_t denotes the price index, and define the h-period average inflation as $\pi_t^h = h^{-1} \sum_{i=0}^{h-1} \pi_{t-i} = \frac{1200}{h} \ln(P_t/P_{t-h})$. Figure 5.1 plots these measures at different horizons. Inflation exhibits substantial time variation, with pronounced spikes during major economic episodes, and displays strong persistence. Consistent with this, the Augmented Dickey-Fuller (ADF) test fails to reject nonstationarity at the 10% level for $h \geq 6$ (see the reported

p-values).

To estimate macroeconomic factors, we use the large-scale FRED-MD dataset of McCracken and Ng (2016). Since inflation typically exhibits persistent "wandering" behavior rather than the explosive exponential growth characteristic of real-activity variables such as GDP and industrial production, we transform variables with explosive trends to stationarity, while retaining moderately nonstationary series (e.g., unemployment and interest rates) in levels to preserve low-frequency variation. After removing series with excessive missing values, the final dataset contains 116 variables over January 1959 to December 2024 (792 monthly observations). Detailed transformations are provided in the online supplement (see online Appendix E).

5.1.2 Forecasting Models

We consider the following forecasting models for $h = 3, 6, 12$ to forecast inflation:

Model 1: TV-FAR-dynamic.

$$\pi_{t+h}^{h} = \theta_0(\tau_t) + \theta_E(\tau_t)'E_t + \theta_G(\tau_t)'G_t + \sum_{i=0}^{3} \theta_{\pi,i}(\tau_t)\Delta\pi_{t-i} + \varepsilon_{t+h}$$

Model 2: TV-FAR-static.

$$\pi_{t+h}^{h} = \theta_0(\tau_t) + \theta_E(\tau_t)'E_t + \sum_{i=0}^{3} \theta_{\pi,i}(\tau_t)\Delta\pi_{t-i} + \varepsilon_{t+h}$$

Model 3: TV-AR.

$$\pi_{t+h}^{h} = \theta_0(\tau_t) + \theta_\pi(\tau_t)\pi_t + \sum_{i=0}^{3} \theta_{\pi,i}(\tau_t)\Delta\pi_{t-i} + \varepsilon_{t+h}$$

Model 4: Semi-TV-FAR-dynamic.

$$\pi_{t+h}^{h} = \theta_0(\tau_t) + \theta_E(\tau_t)'E_t + \theta_G(\tau_t)'G_t + \sum_{i=0}^{3} \theta_{\pi,i}\Delta\pi_{t-i} + \varepsilon_{t+h}$$

Model 5: Semi-TV-FAR-static.

$$\pi_{t+h}^{h} = \theta_0(\tau_t) + \theta_E(\tau_t)'E_t + \sum_{i=0}^{3} \theta_{\pi,i}\Delta\pi_{t-i} + \varepsilon_{t+h}$$

Model 6: FAR-dynamic.

$$\pi_{t+h}^{h} = \theta_0 + \theta_E' E_t + \theta_G' G_t + \sum_{i=0}^{3} \theta_{\pi,i} \Delta\pi_{t-i} + \varepsilon_{t+h}$$

Model 7: FAR-static.

$$\pi_{t+h}^{h} = \theta_0 + \theta_E' E_t + \sum_{i=0}^{3} \theta_{\pi,i} \Delta\pi_{t-i} + \varepsilon_{t+h}$$

Model 8: AR (Benchmark).

$$\pi_{t+h}^{h} = \theta_0 + \theta_\pi \pi_t + \sum_{i=0}^{3} \theta_{\pi,i} \Delta\pi_{t-i} + \varepsilon_{t+h}$$

Models 1 and 2 are time-varying factor-augmented regressions (TV-FAR). Model 1 (TV-FAR-dynamic) includes both nonstationary and stationary factors, whereas Model 2 (TV-FAR-static) includes only nonstationary factors, so their comparison isolates the contribution of stationary factors. Model 3 (TV-AR) is a time-varying autoregressive benchmark used to assess whether latent factors add predictive power beyond parameter instability in observable inflation dynamics.

Models 4 and 5 are semiparametric restricted versions of Models 1 and 2. This comparison provides a robustness check on whether the fully time-varying TV-FAR specifications induce unnecessary flexibility in the short-run autoregressive terms.

Models 6 and 7 are the constant-coefficient counterparts of the dynamic and static FAR specifications, following Hannadige et al. (2024), and Model 8 is the AR benchmark. Comparisons with the AR benchmark measure the overall gains from latent factors and time-varying coefficients, while comparisons among TV-FAR, Semi-TV-FAR, and FAR specifications help separate the roles of time-varying coefficients and factor structure. For all models, the inflation lag length is set to four following Hannadige et al. (2024), and the truncation parameters are selected by modified multifold cross-validation (see the Appendix for details).

Following the IPC_2 criterion of Bai (2004), we extract eight factors from the macroeconomic dataset. The number of identified nonstationary factors varies over time as the data become gradually more stationary as the sample expands, leading to a decline in the number of estimated nonstationary factors over time. Specifically, using the subsample ending in December 1989 (our first initial training window), we estimate six nonstationary factors and two stationary factors, whereas the full-sample estimates yield one nonstationary factor and seven stationary factors. This

variation does not affect our framework, which accommodates predictors with arbitrary persistence, including $I(0)$, $I(1)$, and mixed cases.

To balance parsimony and information content, Models 1, 4, and 6 include all nonstationary factors but only the leading stationary factor. Including additional stationary factors would substantially increase the parameter dimension in the time-varying coefficient setting, given the sieve approximation, potentially leading to overfitting. This specification captures the most dominant cyclical variation while maintaining estimation efficiency. Our focus is to capture dominant cyclical variation; extensions using recently proposed variable selection methods, such as those in Fan et al. (2024); Liu et al. (2025), are left for future work.

5.2 Empirical results

5.2.1 Out-of-sample Forecasting Performance

We employ expanding-window forecasts with initial sample lengths of 30 and 40 years and consider horizons $h = 3, 6, 12$, corresponding to one-quarter-, half-year-, and one-year-ahead forecasts. Forecast accuracy is measured by the mean squared error (MSE) and by its ratio relative to the AR benchmark, where values below (above) one indicate superior (inferior) performance. Table 5.1 reports the relative MSEs of inflation forecasts for the two measures (CPI, PCE) at horizons $h = 3, 6, 12$. Panel A covers the out-of-sample period 1990:01–2024:12, while Panel B focuses on the more recent period 2000:01–2024:12.

The results suggest that the time-varying factor-augmented models compare favorably with their constant-parameter counterparts. Whereas the constant-coefficient FAR specifications, both dynamic and static, frequently produce relative MSEs above one, the TV-FAR specifications yield gains over the AR benchmark in most cases, especially at medium and long horizons. This pattern underscores the importance of accommodating structural instability in the link between macroeconomic factors and inflation. The relative weakness of the TV-AR model in many cases further indicates that allowing for parameter drifts alone is not sufficient: the predictive gains arise from combining factor information with time-varying coefficients. We further decompose the sources of these gains in Section 5.2.3.

A comparison across TV-FAR and Semi-TV-FAR specifications reveals a horizon-dependent pattern with a clear macroeconomic interpretation. At the short horizon ($h = 3$), the static factor specifications perform particularly well: TV-FAR-static delivers the lowest relative MSEs for CPI

Table 5.1: Relative MSEs of inflation forecasts.

	CPI inflation			PCE inflation		
Horizon	3	6	12	3	6	12
Panel A: out-of-sample forecast period 1990:01–2024:12						
TV-FAR-dynamic	1.04	**0.89**	**0.83**	0.99	**0.95**	**0.84**
TV-FAR-static	**0.87**	0.92	1.06	1.00	1.03	1.20
TV-AR	1.03	0.90	0.97	1.02	0.95	1.05
Semi-TV-FAR-dynamic	1.09	0.94	0.90	1.02	1.00	0.86
Semi-TV-FAR-static	0.89	0.95	1.10	**0.96**	1.02	1.16
FAR-dynamic	1.06	1.09	1.09	1.16	1.21	1.24
FAR-static	1.04	1.08	1.14	1.20	1.27	1.36
AR RMSEs (benchmark)	2.58	2.17	1.89	1.85	1.61	1.47
Panel B: out-of-sample forecast period 2000:01–2024:12						
TV-FAR-dynamic	1.02	0.84	0.75	0.96	0.89	**0.80**
TV-FAR-static	**0.82**	**0.77**	**0.70**	**0.87**	**0.81**	0.91
TV-AR	1.04	0.91	0.99	1.04	0.97	1.13
Semi-TV-FAR-dynamic	1.07	0.88	0.78	0.99	0.92	0.80
Semi-TV-FAR-static	0.84	0.81	0.78	0.88	0.88	0.93
FAR-dynamic	1.06	1.09	1.10	1.18	1.22	1.29
FAR-static	1.02	1.03	1.04	1.16	1.19	1.23
AR RMSEs (benchmark)	2.92	2.45	2.11	2.06	1.81	1.63

Note: This table reports the out-of-sample mean squared forecast errors (MSEs) of alternative forecasts of h-period ($h = 3, 6, 12$) average inflation, measured by CPI and PCE, relative to the AR benchmark. The row "AR RMSEs (benchmark)" reports the benchmark RMSEs in levels. Bold entries indicate the best-performing model in each case.

inflation, while the static specifications also perform best for PCE inflation. This is consistent with the view that short-run inflation is governed mainly by its own persistence and by core stochastic trends. As the horizon extends to one year ($h = 12$), the dynamic specifications become more valuable, especially for the longer forecast period starting in 1990 and for PCE inflation. This pattern highlights the role of stationary factors at medium horizons: cyclical shocks, such as output-gap or labor-market pressures, transmit to prices with long and variable lags, so the cyclical signal embedded in G_t today becomes informative about inflation one year ahead.

Nevertheless, in most cases, especially for the dynamic factor specification and at longer horizons, the fully time-varying TV-FAR models remain more accurate than the corresponding Semi-TV-FAR models. Thus, the forecasting results do not suggest that the fully time-varying specification merely overfits redundant short-run dynamics. Rather, full coefficient variation is empirically useful across most settings, while the semiparametric models confirm that the main gains come from time-varying coefficients on the factors rather than from excessive flexibility in the lagged-difference terms.

5.2.2 Linking the Factors to the Phillips Curve

To establish an economic foundation for our forecasting model, we begin with the factor loadings reported in Table 5.2. Panel A shows that the primary nonstationary factor E_t (the single $I(1)$ factor extracted from the full sample) is dominated by nominal interest rates and the price index PCEPI. We therefore interpret it as a *nominal anchor* capturing the long-run stochastic trend of inflation. Panel B shows that the first stationary factor G_t (the $I(0)$ factor in our forecasting model) loads almost exclusively on labor-market and industrial-production indicators, identifying it as a *real-activity* factor that captures cyclical fluctuations. This identification connects naturally to the expectations-augmented Phillips curve, which decomposes inflation into a persistent component driven by inflation expectations π_t^e, a transitory component reflecting the output gap $y_t - y_t^*$, and a supply-side shock ε_t:

$$\pi_t = \pi_t^e + \kappa(y_t - y_t^*) + \varepsilon_t.$$

Table 5.2: Economic interpretation of the latent factors

Rank	Mnemonic	Loading	Description
Panel A: Nonstationary factor (E_t) — nominal/interest-rate anchor			
1	CP3Mx	0.0310	3-Month AA Financial Commercial Paper Rate
2	FEDFUNDS	0.0309	Effective Federal Funds Rate
3	TB6MS	0.0307	6-Month Treasury Bill
4	TB3MS	0.0306	3-Month Treasury Bill
5	GS1	0.0303	1-Year Treasury Rate
6	GS5	0.0280	5-Year Treasury Rate
7	GS10	0.0263	10-Year Treasury Rate
8	PCEPI	0.0262	Personal Cons. Expend.: Chain Index
9	AAAFFM	−0.0262	Moody's Aaa Corporate Bond Minus FEDFUNDS
10	UEMPMEAN	−0.0254	Average Duration of Unemployment (Weeks)
Panel B: Stationary factor (G_t) — real activity/output gap			
1	PAYEMS	−0.7814	All Employees: Total Nonfarm
2	MANEMP	−0.7682	All Employees: Manufacturing
3	IPMANSICS	−0.7573	IP: Manufacturing (SIC)
4	INDPRO	−0.7497	IP Index
5	DMANEMP	−0.7347	All Employees: Manufacturing: Durable Goods
6	IPFPNSS	−0.7229	IP: Final Products and Nonindustrial Supplies
7	IPDMAT	−0.6745	IP: Durable Materials
8	IPFINAL	−0.6737	IP: Final Products (Market Group)
9	IPMAT	−0.6451	IP: Materials
10	IPBUSEQ	−0.6402	IP: Business Equipment

Note: This table lists the 10 variables from the original dataset with the largest absolute loadings on the first nonstationary factor (E_t) and the first stationary factor (G_t) used in our forecasting models.

Our empirical model mirrors this duality by combining $I(1)$ and $I(0)$ factors. The nonstationary factor E_t captures permanent shifts in the nominal anchor and serves as a latent proxy for long-run

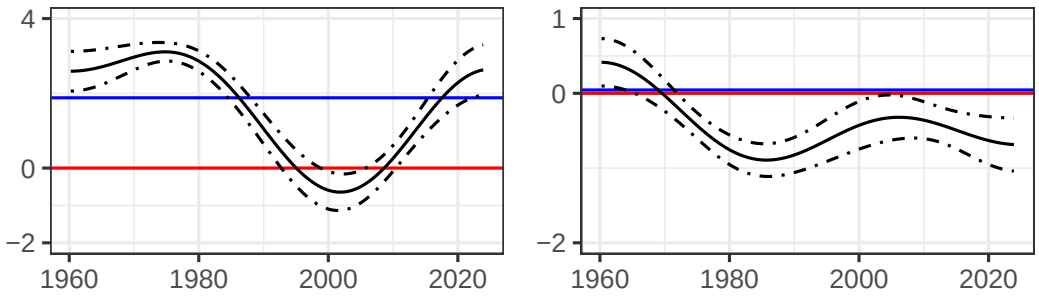
inflation expectations π_t^e, while the stationary factor G_t isolates mean-reverting cyclical pressures and proxies for the output gap $y_t - y_t^*$. Distinguishing between the two stochastic processes allows the model to reflect the dual-frequency nature of inflation dynamics, while supply-side disturbances are absorbed by additional control terms or random errors.

In the TV-FAR-dynamic specification, the time-varying coefficient $\theta_{E,t}$ measures the weight of the long-run nominal trend in current inflation and can be read as an indicator of inflation anchoring. The coefficient $\theta_{G,t}$ corresponds to the slope κ of the empirical Phillips curve (up to the scale implied by the factor loadings), capturing how cyclical slack in the real economy translates into price adjustments. Together, the two coefficients allow us to track how the relative importance of long-run trends and short-run cyclical shocks has evolved over time.

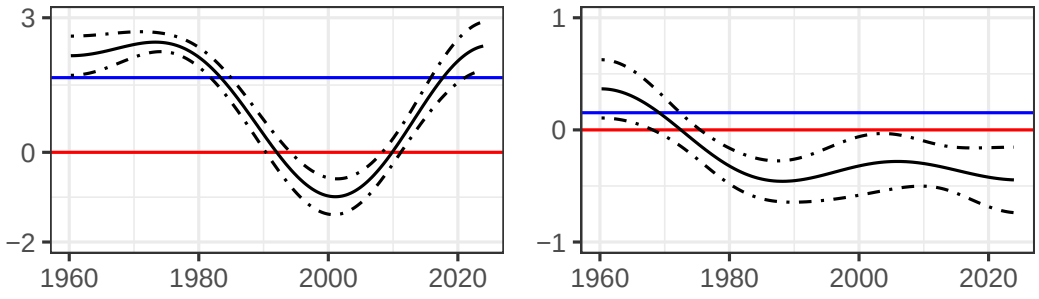
Figure 5.2 further plots the estimated time-varying coefficients (solid black lines) along with their 95% confidence intervals (dashed black lines) for both the nonstationary factor (E_t) and the stationary factor (G_t), based on the full-sample TV-FAR-dynamic model with $h = 12$ (April 1960 to December 2024). For comparison, the constant coefficient estimates from Model 6 (FAR-dynamic) are shown as horizontal blue lines. It can be seen that the nominal-anchor coefficient $\theta_{E,t}$ exhibits a V-shaped trajectory: a high sensitivity of inflation to nominal trends in the 1970s and early 1980s, a sharp decline through the Great Moderation that is consistent with stronger expectation anchoring, and a pronounced resurgence after 2020 that points to a renewed dominance of the nominal trend and a possible risk of de-anchoring. The real-activity coefficient $\theta_{G,t}$ traces the evolution of the Phillips-curve slope. It is negative because the loadings of G_t on real-activity variables are negative; in absolute terms, however, $\theta_{G,t}$ declines steadily from the 1960s to the early 2010s, indicating a sustained flattening of the Phillips curve. More recently, $\theta_{G,t}$ has moved away from zero again, suggesting a re-steepening. This flattening-and-recovery pattern is consistent with the recent evidence in Inoue et al. (2025), yet our latent factor G_t provides a more robust measure by mitigating the measurement errors and idiosyncratic noise inherent in single observable proxies.

5.2.3 Why do factors matter?

Accurately estimating the conditional mean is central to macroeconomic forecasting, both because it provides a forward-looking assessment of economic conditions and because it determines whether forecast errors admit a meaningful economic interpretation. Because macroeconomic



(a) $\theta_E(\tau_t)$ (CPI) (b) $\theta_G(\tau_t)$ (CPI)



(c) $\theta_E(\tau_t)$ (PCE) (d) $\theta_G(\tau_t)$ (PCE)

Figure 5.2: Time-varying coefficient estimates from the TV-FAR-dynamic model ($h = 12$) using the full sample (April 1960 to December 2024).

variables usually display strong autoregressive persistence, a flexible autoregressive specification may already absorb much of the predictable component. A natural question is therefore whether latent factors retain a useful role. Jurado et al. (2015) emphasize that a true forecast error can be identified only after the entire forecastable component has been removed, which requires a sufficiently rich predictive model. Under this view, if a time-varying autoregressive benchmark fails to span the relevant information set, its residuals will continue to contain omitted predictive structure rather than purely unforecastable shocks. Motivated by this observation, we examine whether the latent factors capture predictive components of inflation that the TV-AR benchmark leaves unexplained.

We first decompose the fitted conditional mean from the factor-augmented model. Figure 5.3 reports results from the full-sample TV-FAR-dynamic model with $h = 12$ for CPI inflation, separating the contributions of the time-varying intercept $\widetilde{\theta}(\tau_t)$, the factor component $\widetilde{\theta}^*_F(\tau_t)'\widetilde{F}_t$, and the autoregressive component $\sum_{i=0}^{3} \widetilde{\theta}_{\pi,i}(\tau_t)\Delta\pi_{t-i}$. The factor component accounts for a nontrivial share of the systematic variation in inflation, comoving particularly closely with realized inflation during the prolonged rise and decline of 1960–1980 and around the 2008 financial crisis and the 2020 pandemic recession. As discussed in Remark 1, this evidence suggests that the factor-driven

component captures more than external macroeconomic shocks: by accommodating cointegrating relationships within the system, the latent factors can also proxy for the long-run autoregressive memory of inflation itself. Including nonstationary factors thus provides a broader set of signals for tracking systematic stochastic trends than the historical levels of the target variable alone, and helps to explain why the TV-FAR specification extracts richer predictive content than the TV-AR benchmark.

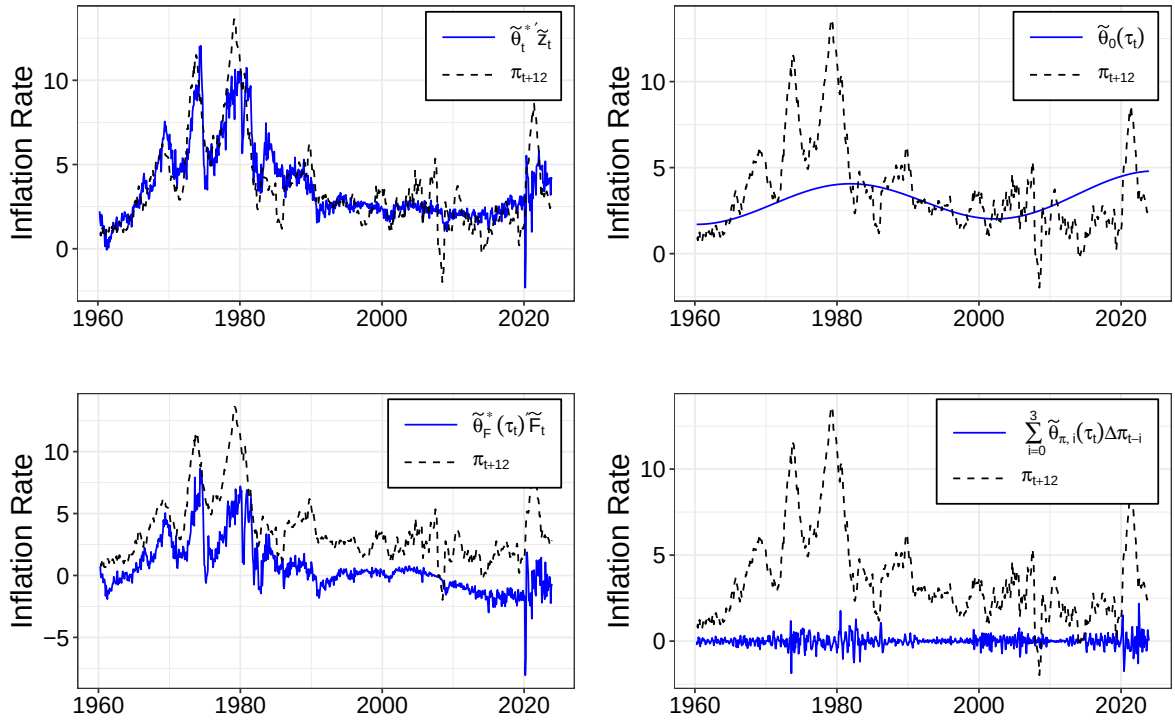


Figure 5.3: Decomposition of the fitted conditional mean from the full-sample TV-FAR-dynamic model ($h = 12$) for CPI inflation.

We next examine whether the residuals from the TV-AR model still contain factor-related predictive information. Figure 5.4 compares the residuals $\widehat{\varepsilon}_{t+12}^{TV\text{-}AR}$ with the extracted $I(1)$ factor $\widetilde{E}_t$ and the first $I(0)$ factor $\widetilde{G}_t$. The residuals exhibit pronounced comovement with both factors over certain periods, indicating that they are not pure noise. To formalize this, we regress $\widehat{\varepsilon}_{t+12}^{TV\text{-}AR}$ on $\widetilde{E}_t$, on $\widetilde{G}_t$, and on both factors jointly,[2] thereby assessing whether the time-t macroeconomic factors carry forward-looking information about the $t+12$ forecast errors of the TV-AR benchmark. Table 5.3 shows that both $\widetilde{E}_t$ and $\widetilde{G}_t$ are significant in the univariate regressions and remain jointly significant when included together. The TV-AR benchmark therefore does not fully span the

[2]This regression is meaningful because the TV-AR residuals are $I(0)$. Under full-sample estimation, $\widetilde{E}_t$ behaves empirically more like a near-integrated process than an exact $I(1)$ series (ADF p-value: 0.202; PP p-value: 0.017; AR(1) coefficient: 0.95). Because the nonstationarity of the factors is much stronger in the early sample, our framework, which accommodates regressors with arbitrary persistence, remains both practically necessary and theoretically robust.

relevant information set, and the latent factors are not redundant even after allowing for flexible parameter dynamics: they capture omitted predictive structure in the residuals and contribute to a more complete approximation of the conditional mean.

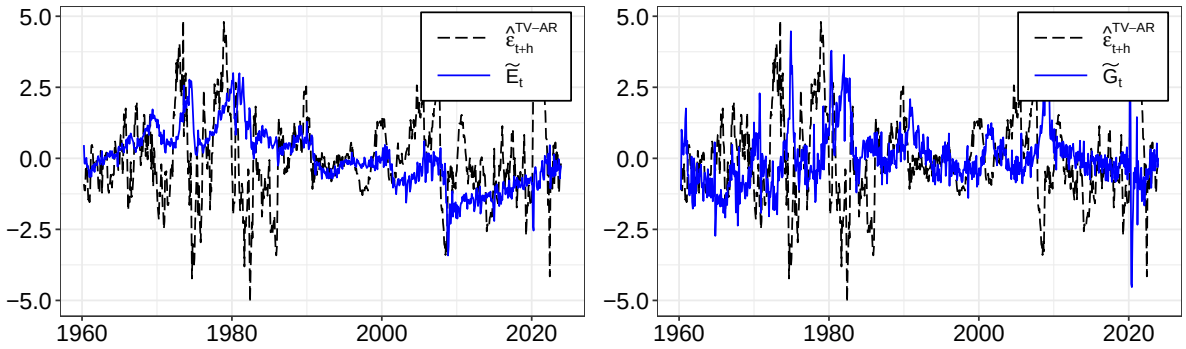


Figure 5.4: Comparison of the residuals from the full-sample TV-AR benchmark for CPI inflation ($h = 12$), $\widehat{\varepsilon}_{t+12}^{TV\text{-}AR}$, with the extracted nonstationary factor $\widetilde{E}_t$ and the first stationary factor $\widetilde{G}_t$.

Table 5.3: Regression of TV-AR residuals on the latent factors

	(1) $\widehat{\varepsilon}_{t+12}^{TV\text{-}AR}$	(2) $\widehat{\varepsilon}_{t+12}^{TV\text{-}AR}$	(3) $\widehat{\varepsilon}_{t+12}^{TV\text{-}AR}$
Constant	−0.0007	−0.0009	−0.0016
	(0.0535)	(0.0507)	(0.0506)
$\widetilde{E}_t$	0.1248**		0.1252**
	(0.0532)		(0.0503)
$\widetilde{G}_t$		−0.4860***	−0.4861***
		(0.0504)	(0.0503)
Observations	765	765	765
R^2	0.0072	0.1085	0.1157
Adjusted R^2	0.0059	0.1073	0.1134
F statistic	5.50**	92.86***	49.85***

Note: This table reports regressions of the residuals from the full-sample TV-AR benchmark for CPI inflation ($h = 12$), $\widehat{\varepsilon}_{t+12}^{TV\text{-}AR}$, on the extracted nonstationary factor $\widetilde{E}_t$, the first stationary factor $\widetilde{G}_t$, and both factors jointly. Standard errors are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Overall, the results highlight the value of our unified forecasting framework, which flexibly accommodates observable and latent predictors, stationary and nonstationary regressors, and smoothly time-varying coefficient functions. The proposed TV-FAR models not only deliver superior out-of-sample performance, but also capture predictive structure left unexplained by the conventional autoregressive benchmark, underscoring the role of latent factors in providing a more complete approximation to the conditional mean.

6 Conclusions

We have introduced a general time-varying factor-augmented forecasting regression designed to extract predictable components under mixed persistence. The framework accommodates observable and latent predictors with stationary, nonstationary, or mixed persistence, allows for both static and dynamic nonstationary factor structures, and nests a broad range of classical and new models as special cases. We have proposed an easy-to-implement sieve-based estimation method for the unknown time-varying coefficients and established large-sample theory for the estimators and prediction intervals. Monte Carlo simulations confirm the good finite-sample performance of the proposed methods. Our empirical analysis of U.S. inflation further shows that latent factors remain informative even in the presence of strong autoregressive dynamics, suggesting that persistence in inflation itself does not exhaust its predictable component. By absorbing latent persistent and cyclical information, the proposed framework provides a more complete approximation to the conditional mean and improves forecasting performance in practice.

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Appendix: Truncation Parameter Selection

An important practical consideration in implementing the series estimation method is the selection of truncation parameters in the orthogonal expansions. The proposed procedure for estimating the time-varying model requires these parameters to be specified *ex ante*. While Generalized Cross-Validation (GCV) has been widely used in stationary contexts (see, e.g., Gao 2007), no comparable data-driven method is readily applicable in nonstationary settings. Given that forecasting performance is a key criterion for model evaluation, we determine the truncation parameters by optimizing out-of-sample forecasting accuracy. Specifically, we adopt a modified multifold cross-validation approach, similar to the criterion proposed by Cai et al. (2000).

Let m and Q be two given positive integers with $T > mQ$. The idea is to use Q validation segments of length m and compute the h-step-ahead forecast errors within each segment. Within each validation segment, the estimation sample is expanded as new observations become available. Let $\widehat{\beta}_{q,t}(k)$ be the estimated coefficients in (2.7) obtained from the subseries $\{(y_s, Z_s, X_s, W_s), 1 \leq s \leq t\}$ with truncation parameter k. The average prediction error based on the qth validation segment is

$$\mathrm{APE}_q(k) = \frac{1}{m} \sum_{t=T-h-qm+1}^{T-h-qm+m} \left(\widehat{y}_{t+h}\big(\widehat{\beta}_{q,t}(k)\big) - y_{t+h} \right)^2,$$

and the overall average prediction error is

$$\mathrm{APE}(k) = Q^{-1} \sum_{q=1}^{Q} \mathrm{APE}_q(k).$$

The data-driven truncation parameter is chosen to minimize $\mathrm{APE}(k)$ over $k \in \mathcal{K}_T$, where $\mathcal{K}_T = \{\max\{2, \lfloor 0.5\, T^{1/5} \rfloor\}, \ldots, \lfloor T^{7/30} \rfloor\}$. In our implementation, we use $m = \lfloor 0.1T \rfloor$ and $Q = 4$, and the truncation parameter is updated by this criterion at each forecasting iteration.